Meeting 6.1 — FreeRTOS Timing Classification & Task Priority Architecture

Project: Digital Harmonium Workstation
Phase: Runtime Architecture (Phase 3)
Section: Meeting 6.1 — Timing Classification, Task Inventory, and Scheduling Philosophy

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1. Objective

Unlike a normal desktop application, an embedded musical instrument cannot simply execute code one function after another.

Many activities occur simultaneously:

- Keyboard scanning
- Audio generation
- LCD updates
- Button reading
- Recording
- Playback
- SD card access
- Background maintenance

Because all of these activities compete for the same CPU, the firmware must define **which activities are more important than others**.

This meeting establishes that hierarchy.

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2. Timing Philosophy

Every software task belongs to one of four timing classes.

Class A — Hard Real-Time

Missing even a single execution deadline produces audible problems.

Examples:

- Audio synthesis
- DMA servicing

These tasks always receive the highest priority.

Class B — Soft Real-Time

Occasional delays are acceptable but should remain extremely small.

Examples:

- Keyboard scanning
- Button scanning
- Voice allocation

The musician may perceive delays if these tasks become excessively late.

Class C — Interactive

These tasks affect usability rather than sound.

Examples:

- LCD updates
- Menu navigation
- Status indicators

A delay of tens of milliseconds is generally acceptable.

Class D — Background

These tasks are not time-critical.

Examples:

- SD card operations
- Diagnostics
- Logging
- File management

These tasks execute only when processor time is available.

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3. Task Priority Hierarchy

The firmware adopts the following priority order.

Priority	Timing Class	Task Category
Highest	Hard Real-Time	Audio Engine
High	Soft Real-Time	Keyboard & Controls
Medium	Interactive	User Interface
Lowest	Background	Storage & Maintenance

This hierarchy ensures that audio playback is never interrupted by user-interface or storage operations.

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4. FreeRTOS Task Inventory

The following tasks form the first version of the firmware scheduler.

Task 1 — Audio Task

Timing Class

Hard Real-Time

Priority

Highest

Purpose

Produces continuous audio samples for the DMA system.

Responsibilities

- Execute synthesis algorithms
- Mix all active voices
- Apply envelopes
- Process filters
- Apply effects
- Fill the next audio buffer

Typical Execution

Runs continuously whenever the instrument is powered.

Failure Consequence

If delayed, audible glitches, clicks, or dropouts occur immediately.

Task 2 — Keyboard Scanner Task

Timing Class

Soft Real-Time

Priority

High

Purpose

Continuously scans the repaired keyboard matrix.

Responsibilities

- Detect new key presses
- Detect key releases
- Perform debouncing
- Generate keyboard events

Typical Scan Rate

Approximately every 1–2 ms.

Failure Consequence

Delayed note response or missed key presses.

Task 3 — Button Scanner Task

Timing Class

Soft Real-Time

Priority

High

Purpose

Monitors dedicated control buttons.

Responsibilities

- Read button matrix
- Debounce switches
- Generate menu events
- Detect long presses
- Detect repeated presses

Typical Scan Rate

Every 5–10 ms.

Failure Consequence

Menus become sluggish or button presses appear inconsistent.

Task 4 — UI Task

Timing Class

Interactive

Priority

Medium

Purpose

Maintains the user interface.

Responsibilities

- Update LCD
- Refresh menus
- Display tempo
- Display transpose
- Display recording status
- Display accompaniment status

Refresh Rate

Approximately 20–30 FPS (30–50 ms per update).

Failure Consequence

Display feels slow, but audio remains unaffected.

Task 5 — LED Manager Task

Timing Class

Interactive

Priority

Medium

Purpose

Controls keyboard LEDs and status indicators.

Responsibilities

- Illuminate pressed keys
- Playback guidance
- Recording guidance
- Status LEDs

Failure Consequence

LED animation becomes less responsive without affecting sound.

Task 6 — Recorder Task

Timing Class

Background

Priority

Low

Purpose

Stores performed note events.

Responsibilities

- Record note-on events
- Record note-off events
- Store timestamps
- Write recording buffers

Failure Consequence

Recording may stop or skip events, but live performance continues.

Task 7 — Playback Task

Timing Class

Soft Real-Time

Priority

Medium-High

Purpose

Replays previously recorded performances.

Responsibilities

- Read recorded events
- Trigger note events
- Synchronize timing
- Drive playback LEDs

Failure Consequence

Playback timing becomes inaccurate.

Task 8 — Storage Task

Timing Class

Background

Priority

Low

Purpose

Handles SD card communication.

Responsibilities

- Load VHL instruments
- Save recordings
- Read recordings
- File management

Failure Consequence

Longer loading times without disturbing live audio.

Task 9 — Diagnostics Task

Timing Class

Background

Priority

Lowest

Purpose

Performs maintenance and debugging activities.

Responsibilities

- Memory statistics
- CPU load monitoring
- Error logging
- Performance counters
- Health checks

Failure Consequence

No effect on instrument operation.

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5. Task Scheduling Philosophy

The scheduler must always preserve the following execution order:

```
Audio Generation
        ↓
Keyboard Input
        ↓
Button Input
        ↓
Playback Timing
        ↓
Display Updates
        ↓
LED Updates
        ↓
Recording
        ↓
Storage
        ↓
Diagnostics
```

Whenever processor time becomes limited, lower-priority tasks are delayed before higher-priority tasks.

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6. Design Rule

The firmware shall never allow:

- LCD updates
- SD card operations
- Recording writes
- Diagnostic calculations

to interrupt audio generation.

Audio playback is always the highest priority activity in the entire system.

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7. Meeting Outcome

This meeting establishes the scheduling philosophy of the Digital Harmonium Workstation.

Nine primary FreeRTOS tasks have been identified and classified into four timing classes:

- Hard Real-Time
- Soft Real-Time
- Interactive
- Background

Each task has a clearly defined purpose, execution priority, and failure consequence.

This hierarchy forms the foundation for all subsequent runtime scheduling, DMA integration, and audio pipeline implementation.

Status: Finalized for Phase 3 Runtime Architecture.